GHULAM ALI GAUHAR

ee21resch01006@iith.ac.in | agauhar066@gmail.com | +91 9030019084 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.12
Sant Kabir College	Class XII (BSEB)	2013	68.20%
High School Rupauli	Class X (BSEB)	2010	70.00%

SKILLS

Languages: C, Python, MATLAB.

TCAD Tools: EDR CAD (In-house TCAD), Sentaurus TCAD.

Plotting Tools: Xmgrace, GNU Plot, Origin.

Operating Systems: Linux, Windows.

PROJECTS

Ultimate Limitation to MOSFET Scaling: A Gate Leakage Perspective:

- Leveraged existing electrostatic framework for modeling gate current in advanced transistors.
- Developed and validated a semi-classical approach to calculate the gate leakage current in multigate MOSFETs.
- Demonstrated that removing/reducing the interfacial SiO2 layer reduces the gate current.
- Proposed optimal device architecture for different channel lengths from the gate current perspective.

Unraveling the Cause of Sub-Thermionic Transport Observed in Schottky Barrier FETs:

- Proposed novel and computationally efficient framework for modeling Schottky barrier FETs.
- Developed and integrated tunneling transport in the in-house TCAD tool.
- Demonstrated transition from step to triangular tunneling barrier as the underlying mechanism for sub-thermionic (SS < 60 mV/dec) transport observed in SBFETs.
- Benchmarked the developed framework by calibrating to multiple experimental data using experimentally observed parameters.
- Designed a Schottky barrier FET-based inverter that achieved 40% reduction in supply voltage over conventional CMOS without degradation in gain.

Modeling of Emerging Non Volatile Memory and Its Application to In-Memory Computing:

- Developed a unified multi-scale simulation platform for modeling of in-Memory computing architecture.
- Integrated kintetic monte carlo based solver for ion and ion-vacancy dynamics with master equation for electron transport.
- Determined upper limit of crossbar-array dimension for performing matrix-vector multiplication.
- Proposed optimized RRAM biasing scheme for program time delay and variability trade off.

PUBLICATIONS

Gate Leakage: "Study of gate current in advanced MOS architectures" in Solid State Electronics (DOI 10.1016/j.sse.2022.108345) [Journal]

Steep Subthreshold Device: "Insights into Sub-Thermionic Transport in Schottky Barrier FETs" in Advanced Theory And Simulations (DOI https://doi.org/10.1002/adts.202400202) [Journal]

Emerging Non-Volatile Memory: "Modeling Framework for In-Memory Computing: From Devices to Crossbar Arrays" in IEEE-ICEE [Conference]

ACHIEVEMENTS

Secured an All India Rank 4078 with 511 GATE Score in GATE 2020 in ECE